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MILITARY EXPENDITURE AND ECONOMIC GROWTH IN DEVELOPING COUNTRIES: EVIDENCE FROM SYSTEM GMM ESTIMATES

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The effect of military expenditure on economic growth in developing countries has been investigated by many empirical literatures. However, there is little consensus of that effect and the diversity seems to come from the use of different models and different estimators. This article applies the Augmented Solow Growth Model to examine the influence of military expenditure on economic growth for 35 developing countries over the period of 1975–2009. By using the system Generalized Method of Moments (GMM) estimators, empirical results indicate that defence has a negative and significant effect on economic growth in the sample countries.

Keywords: Military expenditure; The Augmented Solow Growth Model; Developing countries; System GMM estimates

1. INTRODUCTION

> The single and most massive obstacle to development is the worldwide expenditure on national defence activity.[1]

Military expenditure in developing countries is important and a major aspect of government expenditure which often far exceeds health and education spending. As suggested in the above quotation, many economists believe that military expenditure crowds out more productive expenditure and civilian investment, and thus has a negative impact on economic growth. However, Benoit (1978) found that military expenditure actually accelerates economic growth in less developed countries (LDCs). Since then, many studies have been carried out to assess the impact of military expenditure on economic performances by using different samples, different time periods and different theoretical and empirical meth-

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[1]Quote which was cited in Deger and Sen (1983) is from a statement issued by a UN Committee for Developing Planning in the 1970s.

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ods. However, the results are always controversial and the defence-growth nexus is still important for explicit analysing.

With the end of the Cold War, global military expenditure has been decreasing and it is expected that this reduction in military expenditure (or military burden) will lead to peace dividends for developing countries. But many developing countries still spend a large amount of scarce resources on defence. For example, 12.18% of central government expenditures are spent on defence in lower and middle income countries in 2008 (World Development Indicator, 2011). A lacking of investment and expenditures in education, health and other social-economic activities, many LDCs' economic growth will be affected by high defence spending. For these reasons, the impact of defence spending on economic growth in developing countries needs to be explored and analysed carefully.

The purpose of this article is threefold. First, it aims to examine the influence of military expenditure on economic growth for 35 developing countries in Africa, Latin America and South Asia over the period of 1975–2009, which use longer and newer data series. Secondly, as suggested by Dunne *et al*. (2005), it uses a model which integrates implications stemming from the defence economics literature into the neoclassical growth model. Thirdly, different panel data techniques including dynamic panel data methods are used to estimate the defence–growth relationship, which are believed to provide robust empirical results.

This article is organised as follows: Section 2 provides a brief literature review of the defence–growth nexus studies, especially applying the Solow type growth model. The model, standard cross-section estimation method and dynamic panel methods (the first-differenced GMM and the system GMM estimators) are presented in Sections 3 and 4 gives data and empirical results. Finally, Section 5 presents some conclusions.

2. MILITARY EXPENDIURE AND ECONOMIC GROWTH

Since Benoit's (1978) pioneering work which find that military expenditure has an unexpected positive effect on economic growth for 44 LDCs, various models have been applied to study the defence-growth nexus in developing countries. However, there is neither an agreed theory on the defence-growth nexus nor an empirical consensus on the economic effects of military expenditure. The literature on theoretical models and according empirical studies can be broadly grouped into: *the Benoit-type regressions* (Benoit, 1978; Frederiksen and Looney, 1983; Biswas and Ram, 1986); *supply-side* (Feder-type) *models* (Biswas and Ram, 1986; Mintz and Stevenson, 1995; Batchelor *et al*. 2000; Reitschuler and Loening, 2005; Yildirim *et al*. 2005)), *demand-side models* (Smith, 1980; Faini *et al*. 1984; Deger, 1986b; Knight *et al*. 1996), *demand and supply side models* (Deger and Smith, 1983; Deger and Sen, 1983, 1995; Deger, 1986a, 1986b; Dunne *et al*. 2000; Sezgin, 2001; Galvin, 2003); Klein, 2004); *the Barro models*; (Stroup and Heckelman, 2001; Aizenman and Glick, 2003, 2006; Yakovlev, 2007); *the Solow models*; (Knight *et al*. 1996; Murdoch and Sandler, 2002a, 2002b; Yakovlev 2007).

As for theoretical argues, Benoit's work is only a special case and cannot be reproduced (Biswas and Ram, 1986); The demand-side model has been criticised for the failure to consider supply side issues and are apt to be associated with a negative impact on economic growth; Dunne *et al*. (2005) suggest that 'there seem to be strong theoretical and econometric reasons not to use the Feder-Ram model'; The Deger-type models are always criticised for the ad hoc theoretical specifications; The Barro style growth models are too complex to be estimated explicitly and the theory is just used to suggest variables (Dunne *et al*. 2005).

In general, military expenditure could have negative, positive or insignificant effects on economic growth. For example, Dunne and Uye (2009) survey 102 studies on the economic effects of military expenditure and reveal that almost 39% of the cross-country studies and 35% of the case studies find a negative effect of military expenditure on growth, with only around 20% finding positive for both types of studies. Feder type models reveal that defence spending has no significant or (relative small) positive effect on growth. In contrast, demand side models are associated with a negative effect on growth. The Deger-type models tend to show a net negative effect from defence spending to economic growth in most studies. Thus, the empirical results of the economic effects of military expenditure are mixed which mostly depend on not only sample time periods and countries but also the applied models and methods.

After synthesise and evaluate the above researches undertaken in the area of defence–growth nexus, we may identify information and methodologies that could be relevant to this study. Thus, we focus on the literature which apply popular growth model, i.e. the Solow-type growth model into the analyses of the defence–growth relationship in this article and provide a brief review as follows. Knight *et al*. (1996) extend the standard neoclassical Augmented Solow growth model by incorporating the effect of military spending on growth. The panel estimations for 79 countries during 1971–1985 are carried out and their empirical results indicate that military burden as an explanatory variables has a negative and significant effect on economic growth.

Murdoch and Sandler (2002a, 2002b) use the neoclassical Augmented Solow growth model to empirically test for the influences of a civil war on steady-state income per capita of both home and neighbouring countries. Their empirical analyses reveal that civil wars can have strong negative impacts on economic growth at home and in neighbouring countries. Yakovlev (2007) uses the Augmented Solow growth model to investigate the growth effects of military spending, net arms exports and their interaction and he concludes that military spending and net arms exports have negative impacts on economic growth but their interaction has a positive impact on growth.

Different studies in Table I extend the neoclassical Solow growth model and examined the growth effects of the influencing variables such as military spending, civil war and arms exports. The empirical results of these military related impacts on growth are consis-

TABLE I Review on the Solow Type Literature

Author(s)	Sample	Remarks	Main conclusion
1. Knight *et al.* (1996)	79 countries, 1971–1985	Panel estimation (fixed effect model)	Negative effect of military burden on economic growth
2. Murdoch and Sandler (2002a)	85 countries, 1961–1990	Cross-sectional and panel estimation (fixed effect model)	Civil war created a significant negative impact on short-run growth within the country and its neighbours
3. Murdoch and Sandler (2002a)	31 Africa, 14 Asia, 20 Latin America countries, 1960–1995	Cross-sectional estimation and panel estimation (fixed effect model).	Civil wars can have strong negative impacts on economic growth at home and in neighbouring countries
4. Yakovlev (2007)	28 countries, 1965–2000	Panel estimation (fixed effects, random effects and GMM)	Military spending and net arms exports have negative impacts on economic growth but their interaction has a positive impact on growth

tent and show a negative correlation between defence and growth. Dunne *et al.* (2005) argue that the Augmented Solow model has fewer theoretical weaknesses but is too tight given the range of variables that have been found significant determinants of growth. Furthermore, it might be implausible that the main effect of the military spending is through technology. Military expenditure might influence output in an ad hoc way in the augmented Solow growth model.

3. THE MODEL AND EMPIRICAL ESTIMATORS

3.1. The Augmented Solow Model

Based on the discussion of the economic effect of military expenditure and previous studies (Knight *et al.* 1996), the augmented Solow model with military variable can be parameterised as:

$$\text{growth} = a_0 + a_1 lny_0 + a_2 lnk + a_3 lnh + a_4 ln(n + g + \delta) + a_5 lnm \tag{1}$$

where 'growth' is the growth rate of income per capita in the observational period, Y_0 is the initial level of income per capita, k and h are, respectively investment and human capital proxy, $(n + g + \delta)$ is the growth rate of effective labour plus depreciation and m is military burden. Thus, the military variable is added into the augmented Solow growth model in an ad hoc way.

3.2. Standard Cross-Sectional Regression

By using a cross-section regression framework, Mankiw *et al.* (1992) estimate the long-run economic growth for the period 1960–1985. Similarly, the long-run analysis in this study is based on a regression in the following form:

$$gr_i = \eta + (\alpha - 1) lny_{oi} + \beta_1 lnk_i + \beta_2 lnh_i + \beta_3 ln(n_i + g + \delta) + \beta_4 lnm_i + \epsilon_i \tag{2}$$

where $i = 1, \ldots, 35$ denotes a country index, gr_i is the difference of GDP per capita between 1975 and 2009, and lny_{0i} represents the initial level of GDP per capita of country i. The standard cross-section growth regression suffers from some problems. The main problems are as follows. First, the standard cross-section estimation does not use all available information by reducing the panel data to a time-average data. Second, it suffers from omitted variable bias and measurement errors. The cross-section estimation cannot allow for individual country effects. Third, the endogeneity of one or more regressors is likely to make the cross-section estimation inconsistent. Thus, panel data analysis will be focused in the following empirical studies.

3.3. Dynamic Panel Estimators: the First Difference GMM and the System GMM

The short run estimations are based on 35 countries spanning seven 5-year periods and are analysed by using different panel estimating techniques to judge the robustness of results.

The first-difference GMM, proposed by Arellano and Bond (1991) is introduced into the growth literature by Caselli *et al.* (1996). The dynamic growth regression equation is firstly be taken the first differences to remove unobserved country-specific effects and then in this first-differenced equation, levels of the series lagged two periods or more of the right-

hand-side variables are used as instruments in the regression. By rewriting the above growth equation equivalently, the dynamic growth regression equation is:

$$lny_{it} = \eta_i + \gamma_t + \alpha lny_{i,t-1} + \beta_1 lnk_{it} + \beta_2 lnh_{it} + \beta_3 ln(n_{it} + g + \delta) + \beta_4 lnm_{it} + \epsilon_{it} \quad (3)$$

for $i = 1, \ldots, N$ and $t = 2, \ldots, T$.

Taking the first differences, the first-differenced equation is:

$$lny_{it} - lny_{i,t-1} = \gamma_t - \gamma_{t-1} + \alpha \Delta lny_{i,t-1} + \beta_1 \Delta lnk_{it} + \beta_2 \Delta lnh_{it} + \beta_3 ln(n_{it} + g + \delta) + \beta_4 \Delta lnm_{it} + \Delta\epsilon_{it} \quad (4)$$

for $i = 1, \ldots, N$ and $t = 3, \ldots, T$.

Specially, the following assumptions are made. First, the transient errors are serially uncorrelated (i.e. $E[\varepsilon_{it}\ \varepsilon_{is}] = 0$ for $i = 1, \ldots, N$ and $t \neq s$). Second, the initial conditions satisfy $E[lny_{i1}\varepsilon_{is}] = 0$ for $t \geqslant 2$. Thus, lny_{it-2} and earlier values are correlated with, $\Delta lny_{i,t-1}$ but not with $\Delta\ \varepsilon_{it}$. The values of lny_{it} lagged two periods or more are valid instruments in the first-differenced growth equation.

As in recent empirical growth models, regressors, X_{it} (i.e. the row vector of variables lnk_{it}, lnh_{it}, $ln(n_{it} + n + \delta)$ and lnm_{it}) are all treated as endogenous variables. This means there exist correlations between the current value of regressors and current shocks to lny_{it}, as well as feedback from past shocks to lny_{it} $E[X_{it}\ \varepsilon_{is}] \neq 0$ for $s \leq t$ and $E[X_{it}\ \varepsilon_{is}] = 0$ for $s > t$ are then derived. The moment conditions hold for the first differences equations (Arellano and Bond, 1991) are as follows:

$$E(\Delta\epsilon_{it} y_{it-r} = 0; \quad E(\Delta\epsilon_{it} X_{it-r} = 0) \quad (5)$$

where $r = 2, \ldots, t-1$ and $t = 3, \ldots, T$. So, the values of the endogenous regressors lagged two periods or more are valid instruments as well. In the presence of heteroscedasticity, autocorrelation and endogeneity, Yaffee (2003) recommends that first-differences GMM estimation with robust (the White and Newy-West) panel standard errors is a robust estimator.

However, as shown by Blundell and Bond (1998, 2000) and Bond *et al.* (2001), when the time series are persistent or close to random walk processes, the lagged values of the variables are only weakly correlated with the endogenous variables and are weak instruments. The first-differences GMM estimation also suffers from a loss of valuable observations. Under these conditions, the first-differences GMM estimation is likely to perform poorly and has poor finite sample properties (bias and imprecision). Instead, the system GMM estimator suggested by Arellano and Bover (1995), Blundell and Bond (1998) is more plausible.

The system GMM estimator combines two sets of equations: the standard set of equations in first differences and an additional set of levels equations. The first set of equations is the same as discussed for the first-differences GMM estimation. The other set of levels equations is:

$$lny_{it} = \eta_i + \gamma_t + \alpha\ lny_{i,t-1} + \beta_1 lnk_{it} + \beta_2 lnh_{it} + \beta_3 ln\, n_{it} + g + \delta + \beta_4 lnm_{it} + v_{it} \quad (6)$$

Blundell and Bond (1998) provide the additional assumption that:

$$E(\eta_i \Delta y_{i2}) = 0;\ E(\eta_i \Delta x_{it}) = 0 \quad (7)$$

The sufficient conditions for these additional assumptions are for the series x_{it} and y_{it} to have stationary/time-invariant means. These allow the use of the lagged first differences of

dependent and independent variables as instruments for the level equations. Blundell and Bond (1998) show that the system GMM estimator results in consistent and efficient parameter estimates, and has better asymptotic and finite sample properties than the straightforward first-differences GMM estimator.

As an empirical matter, specification tests proposed by Arellano and Bover (1995) are applied to test the validity of the instruments in our GMM estimation. First, the Arellano–Bond test for the serial correlation is adapted to test whether there is a second-order serial correlation in the first-differenced residuals. The null hypothesis is that the residuals are serially uncorrelated. If the null hypothesis cannot be rejected, it provides the evidence that there is no second-order serial correlation and the GMM estimator is consistent. Second, the Hansen *J*-test and the Diff-in-Hansen test are applied to test the null hypothesis of instrument validity and the validity of the additional moment restriction necessary for system GMM, respectively. Failure to reject this null hypothesis means that the instruments are valid.

Furthermore, we adopt some approaches to improve the efficiency of system GMM estimation. First, according to Roodman (2009b), we restrict the number of instruments used in the system GMM estimation by using only three lags for instruments in the first-differenced equations and collapsing the instrument sets. Second, time-specific effects are included in our growth regression equations to reduce the influence of cross-sectional error dependence in the short dynamic panels (Ding and John, 2011).

4. DATA AND EMPIRICAL RESULTS

4.1. Data

Data on 35 developing countries are derived from several different sources for the period 1975–2009. Data on GDP per capita, investment and population are obtained from the Penn World Table 7.0. Barro and Lee (2010) data set provides a proxy for human capital. Data on military burden are taken from various SIPRI yearbooks and SIPRI online database. Table II display names, description and summary statistics of the variables.

For the empirical analysis, choosing appropriate data for the variables is one of the most important aspects. Each variable has different form of proxies. The data are basically chosen based on specification and availability. Mankiw *et al*. (1992) use per worker GDP and the growth of the workforce. Due to lack of per worker data, per capita data and population growth rate are used in this analysis by following Islam (1995), Caselli *et al*. (1996). Data on investment to GDP ratio are collected to proxy the saving rate. Technological progress pulsing the depreciation rate, $g+\delta$ is assumed to be constant across countries and

TABLE II Variable Descriptions

Variable	Description	Source
lny	Log of real GDP per capita (Laspeyres)	PWT 7.0
lnlagy	Lagged of *lny*	
Growth	The difference of *lny* for the seven 5-year periods	
lnk	Five-year average investment as a share of GDP	PWT 7.0
lnngd	Five-year average population growth rate n plus 0.05	PWT 7.0
lnh	Average number of years of schooling of both sexes 25 years of age or older	Barro-Lee (2010)
m	Five-year average military expenditure as a share of GDP	SIPRI

equals 0.05 (see, Mankiw *et al.* 1992). However, the alternative measure 0.07 is also tried for a robustness check.

Mankiw *et al*. (1992) use school enrolment rates as the proxy of human capital investment. The proxy of the average years of schooling is used instead in this study due to data availability. The data on average years of schooling of both sexes 25 years of age or older are obtained from Barro and Lee (2010) data set. The schooling data at the beginning year of each 5-year period are used. Following by Knight *et al*. (1996), Dunne *et al*. (2005), military expenditure share data are used to measure the effect of military expenditure on growth.

The estimation samples of 35 countries include 16 South American countries, 14 Africa and 5 South Asia countries. Data are available for these 35 countries for all periods and the panel is balanced. In this section, the estimation methods described above are applied to the augmented Solow model with the military variable. The purpose is twofold. First, the growth effect of military expenditure will be examined for the cross-section data and the balanced panel data of 35 developing countries. Second, whether the importance of the econometric issues is borne out by the data will be investigated.

4.2. Empirical Results

4.2.1. Cross-sectional analysis: the long-run estimating results

The long-run estimation results are reported in Table III. The Augmented Solow growth equation is estimated by the Ordinary Least Squares (OLS) estimator with the White heteroscedasticity consistent standard errors and covariance. The estimated coefficient on the initial level of real GDP per capita is significantly negative. Thus, there is an evidence of convergence that poor countries grow faster than rich countries and the convergence rate is approximate 1% a year. The estimated coefficients on *lnngd* and *lnh* are both significant with expected sign. *lnngd* has a negative effect on growth and *lnh* has a positive effect on growth. The estimated coefficient on *lnk* is insignificant. The growth effect of military expenditure is negative but insignificant. Thus, the estimation results are not strong enough to make the conclusion of the negative effect from defence to economic growth in the long run.

4.2.2. Panel estimating results

The panel estimation results are reported in Table IV. The econometrics package used is Stata 10. The first three columns of Table IV report the results using the OLS levels, Fixed Effects Model (FEM) and Feasible Generalized Least Squares (FGLS) estimators, respectively. The fourth column presents the results using the first-differences GMM estimator and the last column reports from using the system GMM estimators. Both GMM estimators are two-step which are asymptotically more efficient than one-step. However, the

TABLE III The Long-Run Estimating Results

Variable	Coefficient	Std. error	*t*-Statistic	Prob.
Constant	2.495**	1.134	2.200	0.036
lnY75	−0.303***	0.093	−3.242	0.003
lnk	0.066	0.283	0.233	0.817
lnngd	−0.594**	0.217	−2.740	0.010
lnh	0.324*	0.167	1.935	0.063
lnm	−0.010	0.154	−0.062	0.951
R^2	0.367	Prob(F-statistic)	0.016	

Notes: *lnY75* is the log of real GDP per capita in the initial year, 1975.
*at 10%; **at 5%; and ***at 1%.

TABLE IV The Effect of Military Expenditure in the Augmented Solow Growth Model

	OLS	FEM	FGLS	DIF-GMM	SYS-GMM
lnlagy	−0.027**	−0.379***	−0.042***	-0.287***	−0.058***
	(−0.012)	(0.044)	(0.008)	(0.224)	(0.036)
lnk	0.039**	0.124***	0.037**	0.093	0.154**
	(0.019)	(0.031)	(0.018)	(0.060)	(0.075)
lnngd	−0.068***	−0.027	−0.060***	−0.009	−0.024
	(0.019)	(0.027)	(0.015)	(0.056)	(0.052)
lnh	0.016	−0.179***	0.022	−0.043	0.104
	(0.020)	(0. 058)	(0.017)	(0.202)	(0.077)
lnm	−0.019	−0.003	−0.024***	−0.091	−0.090*
	(0.012)	(0.019)	(0.009)	(0.069)	(0.047)
Constant	0.173*	3.067***	0.327***		
	(0.105)	(0.371)	(0.084)		
Instruments				75	39
Hansen *J*-test				[1.000]	[0.545]
Diff-in-Hansen test					[0.559]
AR(1)				[0.021]	[0.000]
AR(2)				[0.434]	[0.369]

Notes:
Dependent variable is *growth.* All time dummies are included but not reported to save space.
Standard errors are shown in parentheses, *p*-values in brackets.
DIF-GMM' is the first-differences GMM estimate and 'SYS-GMM' is the system GMM estimate. Syntax used to obtain estimates of GMM is xtabond2 in Stata (Roodman, 2009a). Both GMM regressions are two-step and the Windmeijer (2005) finite sample correction for standard errors is employed.
Instrument used for DIF-GMM is lagged two and further periods of $\ln y_{it}$ $\ln k_{it}$ $\ln h_{it}$ $\ln(n_{it} + n + \delta)$, and $\ln m_{it}$ Following Roodman (2009b), instrument matrix is collapsed and only use three lags as instruments in SYS-GMM regression.
The values reported for the Hansen *J*-test and the Diff-in-Hansen test are the *p*-values for the null hypothesis of instrument validity and the *p*-values for the validity of the additional moment restriction necessary for system GMM, respectively. The values reported for AR(1) and AR(2) are the *p*-values for first-and second-orderauto-correlated disturbances in the first differences equations.
*at 10%; **at 5%; and ***at 1%.

two-step standard errors tend to be severely downward biased (Arellano and Bond, 1991; Blundell and Bond, 1998). Hence, the Windmeijer (2005) finite sample correction for standard errors is employed to make two-step robust estimations more efficient. Period dummies are found to be jointly significant in every regression. In order to conserve space, the coefficients on the period dummies are not presented.

As mentioned by Bond *et al*. (2001), omitting variables (i.e. unobserved country-specific effects) will give an estimate of the coefficient on *lnlagy* which is biased upward. The FEM will cause an estimate of this coefficient to be seriously downward biased. Thus, the estimated coefficient on *lnlagy* from OLS and FEM can be regarded as an approximate upper bound and lower bound, respectively. A consistent estimate of the coefficient can be expected to lie in these two bounds. This simple indication is useful to detect the first-differences GMM results. If the first-differences GMM estimate lies close to or below the FEM estimator, it is also biased downward due to weak instruments.

Beginning with OLS results, the estimated coefficients on *lnlay, lnk* and *lnngd* are statistically significant and with the expected sign. The estimated coefficient on *lnh* is positive but insignificant. The estimated coefficient on *lnm* is negatively but insignificantly related to economic growth. Then when a FEM estimator is used, the results show that military burden is insignificant. The estimated coefficients on lagged GDP per capita and investment are significant with the expected sign. However, the estimated coefficient on *lnh* is significant with wrong sign. Column 3 presents the results using the two-way fixed effects FGLS estimator. It allows the estimation in the presence of panel specific Autoregressive

(AR) (1) autocorrelation and heteroscedasticity across panels. All estimated coefficients except human capital variable are significant and with the expected signs. The estimated coefficient on *lnlagy* falls between the upper and lower bounds and military burden has a significant and negative effect on growth.

Using the first-differences GMM estimator, the results of the Arellano-Bond tests indicate that there is no second-order serial correlation and the results of Hansen *J*-test indicate that the instruments are valid. Only the estimated coefficients on initial income are significant and the estimate of the estimated coefficient on human capital measure has an unexpected negative sign. So the first-differences GMM estimator is likely to be poorly behaved.

The last column of Table IV presents the system GMM estimate. The results of the Arellano-Bond tests indicate that there is no second-order serial correlation. We do not reject the null hypothesis of the Hansen *J*-test and the Diff-in-Hansen test which indicate the test statistics show a proper specification. Here the estimate of the coefficient on *lnlagy* (-0.058) is significant and lies above the corresponding FEM estimate and below the corresponding OLS estimate. Physical investment has a significant and positive effect on growth. Both the growth rate of population and the variable of schooling years as the measure of human capital are not significant determinants of economic growth. Military burden had a significant and negative effect on growth.

Using the OLS levels, the FEM (Least Squares Dummy Variables (LSDV)), the FGLS, the first differences and system GMM estimators, the effect of military burden on economic growth in Augmented Solow model is investigated. The estimated results are summarised as follows. First, military burden has a negative and significant effect on economic growth in the robust estimations (i.e. the System GMM). It implies that after controlling for investment, population growth and human capital measurement, military burden exerts a direct negative impact on growth in 35 developing countries.

Second, the estimate coefficients on the initial level of income are significant and negatively related to economic growth by all estimators used above. This suggests the existence of 'conditional convergence' and is consistent with the standard results of the empirical growth literature. The convergence rate is only approximate 1% a year by the System GMM which is similar with the finding of Mankiw *et al*. (2002) and our long-run convergence rate. However, as Nerlove (2000) has mentioned, the measuring of convergence rate depends on the different choice of estimator. An uncertainty is acknowledged in this measurement result. Third, the comparison of both different estimators and the estimate results shows the importance of the choice of robust estimators. The two-step System GMM estimator is a robust estimator in this study and produce reasonable and compatible empirical results

5. CONCLUSION

The defence–growth nexus has been an issue of keen concern in defence economics and there is a large amount of the literature investigating the growth effect of military expenditure in developing countries. However, the existing literature is inconclusive as to the effect of defence on economic growth due to applying different theoretical models, different empirical techniques and different samples. After reviewing some defence-growth literature which integrate military variables into the augmented Solow growth model, this paper examines the influence of defence on economic growth in 35 developing countries. Our panel regressions present reasonable and robust results by applying more recent econometric techniques such as the dynamic panel System GMM estimators. The empirical

panel results indicate that defence has a significant and negative effect on economic growth in 35 sample developing countries.

Thus, the empirical estimations support the negative effect of defence on growth and are consensus of Deger and Sen's (1983) opinion that 'the single and most massive obstacle to development is the worldwide expenditure on national defence activity'. Furthermore, proper regression model and more advanced econometric methodologies do improve empirical results in this article which could make contributions to the defence economics literature.

References

Aizenman, J. and Glick, R. (2003) *Military Expenditure, Threats and Growth, NBER Working Paper, 9618*. National Bureau of Economic Research, 1050 Massachusetts Avenue Cambridge, MA 02138, U.S.A.

Aizenman, J. and Glick, R. (2006) Military expenditure, threats and growth. *The Journal Of International Trade & Economic Development: An International And Comparative Review* **15**(2) 129–155.

Arellano, M. and Bond, S. (1991) Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *Review of Economic Studies* **58** 277–297.

Arellano, M. and Bover, O. (1995) Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics* **68** 29–52.

Barro, R.J. and Lee J.W. (2010) *A New Data Set of Educational Attainment in the World, 1950–2010, NBER Working Paper No. 15902*. National Bureau of Economic Research, 1050 Massachusetts Avenue Cambridge, MA 02138, U.S.A.

Batchelor, P., Dunne, P. and Saal, D. (2000) Military spending and economic growth in South Africa. *Defence and Peace Economics* **11**(6) 553–571.

Benoit, E. (1978) Growth and defence in LDCs. *Economic Development and Cultural Change* **26** 271–280.

Biswas, B. and Ram, R. (1986) Military spending and economic growth in less developed countries: an augmented model and further evidence. *Economic Development and Cultural Change* **34**(2) 361–372.

Blundell, R. and Bond, S. (1998) Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics* **87**(1) 115–143.

Blundell, R. and Bond, S. (2000) GMM estimation with persistent panel data: an application to production functions. *Econometric Reviews* **19**(3) 321–340.

Bond, S., Hoeffler, A. and Temple, J. (2001) *GMM Estimation of Empirical Growth Models, No 2001–W21*. Oxford, UK: Economics Group, Nuffield College, University of Oxford.

Caselli, F., Esquivel, G. and Lefort, F. (1996) Reopening the convergence debate: a new look at cross-country growth empirics. *Journal of Economic Growth* **1** 363–389.

Deger, S. (1986a) Economic development and defence expenditure. *Economic Development and Cultural Change* **35**(1) 179–196.

Deger, S. (1986b) *Military Expenditure and Third World Countries: The Economic Effect*. London: Routledge and Kegan Paul.

Deger, S. and Sen, S. (1983) Military expenditure, spin-off and economic development. *Journal of Development Economics* **13** 67–83.

Deger, S. and Sen, S. (1995) Military expenditure and third world countries. In *Handbook of Defence Economics*, 1, edited by K. Hartley and T. Sandler. Amsterdam: Elsevier, 275–307.

Deger, S. and Smith, R. (1983) Military expenditure and growth in less developed countries. *Journal of Conflict Resolution* **27**(2) 335–353.

Ding, S. and John, K. (2011) Why has China grown so fast? the role of physical and human capital formation *Oxford Bulletin of Economics and Statistics* **73**(2) 141–174.

Ding, S. and Knight, J. (2011) Why has China grown so fast? the role of physical and human capital formation *Oxford Bulletin of Economics and Statistics* **73**(2) 141–174.

Dunne, J.P., Nikolaidou, E. and Roux, A. (2000) Defence spending and economic growth in south africa: a supply and demand model. *Defence and Peace Economics* **11**(4) 573–585.

Dunne, J.P., Smith, R.P. and Willenbockel, D. (2005) Models of military expenditure and growth: a critical review. *Defence and Peace Economics* **16**(6) 449–461.

Dunne, P. and Uye, M. (2009) Military Spending and Development. In The Global Arms Trade, pp. 293–305 edited by A. Tan. London: Europa/Routledge.

Faini, R., Annez, P. and Taylor, T. (1984) Defence spending. Economic structure and growth evidence among countries and overtime. *Economic Development and Cultural Change* **32**(3) 487–498.

Frederiksen, P.C. and Looney, R.E. (1983) Defence expenditures and economic growth in developing countries. *Armed Forces and Society* **9**(4) 633–645.

Galvin, H. (2003) The Impact of defence spending on the economic growth of developing countries: a cross-section study. *Defence and Peace Economics* **14**(1) 51–59.

Islam, N. (1995) Growth empirics: a panel data approach. *Quarterly Journal of Economics* **110**(4) 1127–1170.

Klein, T. (2004) Military Expenditure and Economic Growth: Peru 1970–1996. *Defence and Peace Economics* **15** (3) 275–288.

Knight, M., Loayza, N. and Villanueva, D. (1996) The peace dividend: military spending cuts and economic growth. *IMF Staff Papers* 1–44.

Mankiw, N.G., Romer, D. and Weil, D.N. (1992) A contribution to the empirics of economic growth. *Quarterly Journal of Economics* **107** 407–437.

Mintz, A. and Stevenson, R.T. (1995) Defence expenditures, economic growth, and the 'peace dividend': a longitudinal analysis of 103 countries. *Journal of Conflict Resolution* **39** 283–305.

Murdoch, J.C. and Sandler, T. (2002a) Economic growth, civil wars, and spatial spillovers. *Journal of Conflict Resolution* **46**(1) 91–110.

Murdoch, J.C. and Sandler, T. (2002b) Civil wars and economic growth: a regional comparison. *Defence and Peace Economics* **13**(6) 451–464.

Nerlove, M. (2000) Growth rate convergence, fact or artifact? an essay on panel data econometrics. In *Panel Data Econometrics: Future Directions: Papers in Honour of Professor Pietro Balestra,* edited by E. Ronchetti. Amsterdam: North-Holland, pp. 3–34.

Penn World Table 7.0, Centre for International Comparisons of Production, Income and Prices (CIC), University of Pennsylvania. http://pwt.econ.upenn.edu/php_site/pwt_index.php

Reitschuler, G. and Loening, J.L. (2005) Modeling the Defense-Growth Nexus in Guatemala. *World Development* **33**(3) 513–526.

Roodman, D. (2009a) How to do xtabond2: an introduction to difference and system GMM in Stata. *Stata Journal* **9** 86–136.

Roodman, D. (2009b) A note on the theme of too many instruments. *Oxford Bulletin of Economics and Statistics* **71** 135–158.

Sezgin, S. (2001) An empirical analysis of Turkey's defence–growth relationships with a multi-equation model (1956–1994). *Defence and Peace Economics* **12**(1) 69–86.

SIPRI (1982, 1989, 1990, 2011) *SIPRI Yearbook. World Armaments and Disarmaments*. Oxford: Oxford University Press.

Smith, R. (1980) Military expenditure and investment in OECD countries 1954–1973. *Journal of Comparative Economics* **4**(1) 19–32.

Stroup, M.D. and Heckelman, J.C. (2001) Size of the military sector and economic growth: a panel data analysis of Africa and Latin America. *Journal of Applied Economics, IV* **2** 329–360.

Windmeijer, F. (2005) A finite sample correction for the variance of linear efficient two-step GMM estimators. *Journal of Econometrics* **126** 25–51.

World Development Indicators. Retrieved 2011, from World Bank Database: http://esdsw2.mc.manchester.ac.uk/WDS_WB/TableViewer/dimView.aspx?ReportId=57

Yaffee, R. (2003) *Connect: Information Technology*,.www.nyu.edu/its/socsci/Docs/pda.pdf

Yakovlev, P. (2007) Arms trade, military spending, and economic growth. *Defence and Peace Economics* **18**(4) 317–338.

Yildirim, J., Sezgin, S. and Ocal, N. (2005) Military expenditure and economic growth in middle eastern countries: a dynamic panel data analysis. *Defence and Peace Economics* **16**(4) 283–295.